# Lab: Senses and Signals -- How Your Body Gathers Information

## Objective

> **Learning Goal:** Investigate how your five senses (plus internal senses) collect data for your brain, and discover how your brain combines and predicts sensory information.

This is an **investigation lab** designed for middle school students (Grades 6-8).

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## Materials Needed

- Blindfold or eye mask
- Cotton balls with different scents (vanilla, lemon, mint)
- Small food samples for taste test (optional)
- Ice cube or cold pack
- Lab journal or notebook

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## Part 1: Sensory Inventory (10 min)

You have more than five senses. Besides sight, hearing, touch, taste, and smell, your body also has internal senses: balance (vestibular), body position (proprioception), temperature, pain, and hunger.

| Sense | Sensor Organ/ Location | What does it detect? | Example of a prediction it makes |
|---|---|---|---|
| Sight | Eyes | Light, color, movement | |
| Hearing | Ears | Sound waves | |
| Touch | Skin | Pressure, texture | |
| Taste | Tongue | Chemicals in food | |
| Smell | Nose | Chemicals in air | |
| Balance | Inner ear | Body orientation | |
| Proprioception | Muscles and joints | Body position | |
| Temperature | Skin and hypothalamus | Hot and cold | |

*Write your response here*

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## Part 2: Sensory Prediction Experiments (15 min)

Your brain does not just receive sensory data -- it PREDICTS what the senses will report. Test this with these mini-experiments.

**Experiment A -- Temperature Illusion:** Hold an ice cube in one hand for 30 seconds, then touch a room-temperature surface with both hands. The warm hand feels normal. What does the cold hand feel?

**Experiment B -- Smell Adaptation:** Hold a scented cotton ball near your nose for 60 seconds. Rate the smell intensity every 15 seconds.

| Time | Smell Intensity (1-10) |
|---|---|
| 0 sec | |
| 15 sec | |
| 30 sec | |
| 45 sec | |
| 60 sec | |

**Experiment C -- Visual Prediction:** Stare at a fixed point for 30 seconds, then quickly look at a blank wall. Do you see an afterimage?

*Write your response here*

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## Part 3: Sensory Integration (10 min)

Your brain combines information from multiple senses to build one unified experience. When senses conflict, interesting things happen.

**The McGurk Effect (if video available):** Watch a video where the mouth shapes do not match the sound. What do you hear?

**Discuss with your partner:**

| Question | Your Answer |
| --- | --- |
| When have your senses disagreed with each other? | |
| What happens in your body during motion sickness? | |
| Why is food less tasty when you have a cold? | |
| How do noise-canceling headphones change your perception? | |

*Write your response here*

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## Part 4: Reflection Table

| Question | Your Answer |
| --- | --- |
| How many senses do you actually have? | |
| What does sensory adaptation tell you about prediction? | |
| Why does your brain combine multiple senses? | |
| What is a sensory prediction error? Give a body example. | |
| How does Active Inference explain perception in the body? | |

*Write your response here*